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Event-Based Menu Recommendation in New York Using Sentiment Analysis of Food Preferences

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Declaration

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Abstract

In the competitive New York City culinary market, catering managers traditionally rely on intuition for menu design, which often leads to a disconnect with shifting public preferences. This project addresses the need for data-driven insights by bridging the gap between available recipe data and public perception through a context-aware recommendation framework. The proposed system utilizes a Medallion Architecture—comprising Bronze, Silver, and Gold layers—to organize the flow from raw data ingestion to refined analysis. The methodology integrates structured recipe metadata from the Spoonacular API with unstructured editorial discourse from the Eater NY RSS feed using Natural Language Processing techniques like VADER and spaCy. Results from the initial implementation phase demonstrate the successful ingestion of a 40-record sample into the Bronze layer with an overall ingestion efficiency of 80Validation of these raw records confirmed the presence of critical metadata for dietary filtering, such as vegetarian and gluten-free attributes, while highlighting necessary preprocessing tasks like HTML sanitization. These findings demonstrate the feasibility of a data-driven menu recommendation approach that synthesizes both factual and perceptual culinary dimensions. Ultimately, the project establishes a scalable foundation for future sentiment-driven analytics, enabling professional event planners to align menus more effectively with real-time urban culinary trends.

Keywords: Menu Recommendation, Sentiment Analysis, Natural Language Processing, Data Engineering.

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Contents

List of Tables v

1 Introduction 1
- 1.1 Background . . . 1
- 1.2 Problem Statement . . . 1
- 1.3 Objectives . . . 1
- 1.4 Scope . . . 2
- 1.5 Assumptions . . . 2
- 1.6 Limitations . . . 2
- 1.7 Report Structure . . . 2

2 Literature Review 3
- 2.1 Overview . . . 3
- 2.2 Sentiment Analysis and NLP in the Food Domain . . . 3
- 2.3 Context-Aware and Event-Based Recommendation . . . 3
- 2.4 Summary and Research Gaps . . . 4

3 Methodology 5
- 3.1 Overview . . . 5
- 3.2 Data Acquisition and Integration . . . 5
- 3.3 Data Processing and NLP Sub-pipeline . . . 5
- 3.4 Orchestration Strategy . . . 6
- 3.5 Aggregation and Visualization . . . 6

4 Implementation and Results 7
- 4.1 Overview . . . 7
- 4.2 Implementation Details . . . 7
- 4.3 System Features and Interface . . . 7
- 4.4 Experimental Setup . . . 7
- 4.5 Results . . . 8
- 4.6 Validation results . . . 8

5 Discussion 9
- 5.1 Overview . . . 9
- 5.2 Interpretation of Results . . . 9
- 5.3 Addressing Project Objectives . . . 9
- 5.4 Limitations and Scalability . . . 10
- 5.5 Comparison with Related Work . . . 10
- 5.6 Potential Improvements . . . 10

6 Conclusions **11**
6.1 Summary 11
6.2 Conclusions 11
6.3 Recommendations for Future Work 11

References **13**

List of Tables

4.1 Summary of Performance Results (Ingestion Phase) 8

List of Abbreviations

ABSA	Aspect-Based Sentiment Analysis
API	Application Programming Interface
CARS	Context-Aware Recommender Systems
DAG	Directed Acyclic Graph (Airflow)
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
NER	Named Entity Recognition
NLP	Natural Language Processing
RSS	Really Simple Syndication
SMPCS	School of Mathematical, Physical and Computational Sciences
TF-IDF	Term Frequency-Inverse Document Frequency
UoR	University of Reading
VADER	Valence Aware Dictionary and Sentiment Reasoner
XML	Extensible Markup Language

Chapter 1

Introduction

1.1 Background

New York City represents one of the most competitive and gastronomically diverse environments globally, where catering managers and event planners must regularly curate menus for events ranging from corporate dinners to weddings. In this high-stakes environment, selecting the appropriate dishes to match specific audience preferences is critical for both commercial and strategic success. Traditionally, these menu design decisions have relied heavily on intuition and prior experience rather than on a systematic analysis of current food trends and public opinion. While online platforms like food blogs, reviews, and culinary forums offer a wealth of information regarding dining preferences, this data remains largely underutilized for menu planning. Advances in Natural Language Processing (NLP) and data engineering now provide the tools necessary to bridge the gap between available dish data and the public's perception of those dishes.

1.2 Problem Statement

The primary problem addressed by this project is that menu design decisions are often made without data-driven insights, leading to a disconnect between offered menus and shifting culinary trends. Event planners lack a streamlined method to transform heterogeneous and unstructured data sources—such as editorial RSS feeds and recipe APIs—into actionable insights. Furthermore, extracting meaningful sentiment from informal and noisy textual data presents significant technical challenges that hinder the ability to make informed menu selections based on real-time public preferences.

1.3 Objectives

The objectives of this project are focused on fulfilling specific user needs for catering managers and event planners in New York City The objectives of this project are:

1. View and identify the highest-rated dishes for specific event types, such as corporate gatherings or weddings.
2. Filter dish recommendations according to diverse dietary restrictions, including vegan, gluten-free, and halal requirements.
3. Visualize insights through an interactive dashboard to support weekly or per-event decision-making.

1.4 Scope

This section describes the boundaries of the project. The following topics are **within scope**:

- Integration of structured culinary data from the Spoonacular API.
- Extraction of unstructured public discourse from the Eater NY RSS feed.
- Pipeline orchestration using Apache Airflow and visualization via Plotly Dash

The following topics are **out of scope**:

- Automated data collection from platforms like TripAdvisor, which has a restricted terms of service.
- The system is designed for batch processing on a weekly schedule; streaming data or "live" social media sentiment (e.g., from X/Twitter) is not supported.
- The project focuses on descriptive and diagnostic analysis of current and past data rather than using machine learning to predict future culinary trends.
- Data collection and sentiment analysis are restricted to the New York City metropolitan area; regional trends in other cities or international markets are not considered.

1.5 Assumptions

The following assumptions were made during the development of this project:

- The Spoonacular API serves as a reliable source for well-formed and accurate recipe and ingredient data.
- The Eater NY RSS feed provides a sufficient proxy for current culinary trends and public discourse in the New York City area.
- A weekly update schedule for data ingestion is sufficient because food trends in established urban environments evolve relatively slowly

1.6 Limitations

The following limitations apply to this project:

- The system requires significant preprocessing to handle HTML entities, boilerplate noise, and complex nested JSON structures from the data sources.
- Initial validation was performed on limited record samples, though the architecture is designed for scalability
- The use of RSS feeds reflects editorial bias rather than direct, unfiltered user opinions.

1.7 Report Structure

The remainder of this report is structured as follows: Chapter 2 presents the literature review. Chapter 3 describes the methodology. Chapter 4 presents the implementation and results. Chapter 5 discusses the findings. Chapter 6 concludes the report.

Chapter 2

Literature Review

2.1 Overview

This chapter explores the foundational research and technological advancements that underpin event-based menu recommendation and culinary sentiment analysis. Traditional recommender systems have historically relied on data-centric descriptors, such as explicit ratings and user metadata, to predict preferences. However, the growing field of affective computing and context-aware recommender systems (CARS) has demonstrated that user preferences are highly situational and influenced by transient states like emotions, moods, and environmental context. While significant work has been done in general sentiment analysis and recipe retrieval, there remains a need for systems that synthesize heterogeneous data sources—combining structured culinary facts with unstructured public discourse—to support professional menu planning for specific events

2.2 Sentiment Analysis and NLP in the Food Domain

Extracting meaningful insights from food-related text requires a transition from basic keyword matching to granular Opinion Mining.

- Methodologies: Research has evolved from dictionary-based approaches using sentiment lexicons like SentiWordNet to sophisticated deep learning architectures such as BERT and RoBERTa, which achieve high accuracy in capturing complex semantic features
- Culinary Specifics: In the food industry, sentiment analysis is used to decode online consumer feedback regarding taste, texture, and pricing. Recent studies have highlighted the effectiveness of Aspect-Based Sentiment Analysis (ABSA), which links specific attributes (e.g., a specific dish) to its perceived polarity, significantly improving recommendation precision over simple rating-based models
- Informal Text Processing: For the short, noisy, and informal text common in web discourse, the VADER model is recognized for its low computational cost and high effectiveness in quantifying opinion polarity

2.3 Context-Aware and Event-Based Recommendation

User choices are rarely static; they are "reference events" influenced by the "when," "where," and "how" of interactions.

- Integration Paradigms: Literature identifies three primary ways to integrate context: pre-filtering (filtering data before recommending), post-filtering (re-ranking recommendations after initial computation), and contextual modeling (integrating context directly into the algorithm). While pre-filtering is easy to implement using traditional 2D models, contextual modeling via Factorization Machines often captures the multidimensional nature of interactions more accurate.
- The Consumption Chain: Emotions and situational context play different roles at various stages: the entry stage (initial mood), the consumption stage (experience during the event), and the exit stage (satisfaction feedback)
- Event Drivers: In New York City's competitive environment, social factors—such as group size and occasion type (e.g., weddings vs. corporate dinners)—act as primary triggers that can override long-term biological food preferences

2.4 Summary and Research Gaps

The literature reviewed highlights several critical gaps that this project aims to address:

- Event-Specific Menu Personalization: Most existing food recommender systems focus on individual users or generic health advice rather than providing actionable insights for professional catering managers based on shifting urban culinary trends and event contexts.
- Integration of Heterogeneous Data: While systems exist for recipe retrieval and separate sentiment analysis, there is a lack of end-to-end models that bridge the gap between structured recipe data (APIs) and real-time unstructured public trends (RSS feeds) specifically for menu curation.

Chapter 3

Methodology

3.1 Overview

The system implements a data lake Architecture, which organizes data into three progressive layers—Bronze, Silver, and Gold—to ensure logical separation between raw ingestion and refined analysis. This structure allows for a scalable and traceable data flow, transforming heterogeneous inputs into actionable insights.

3.2 Data Acquisition and Integration

The approach integrates two complementary data sources to capture both factual and perceptual dimensions of food data

- Structured Data (Spoonacular API): The system utilizes the /recipes/findByIngredients REST endpoint to retrieve detailed recipe titles, identifiers, and ingredient metadata in JSON format. This source allows for precise filtering based on dietary restrictions and intolerances, ensuring recommendations meet specific guest requirements
- Unstructured Data (Eater NY RSS Feed): To capture real-time culinary trends in New York City, the system ingests XML article summaries. This source provides descriptive language about restaurant experiences and food trends, serving as the primary input for sentiment and trend analysis. Unlike web-scraping restricted sites, the RSS format ensures stable and ethical data extraction

3.3 Data Processing and NLP Sub-pipeline

Once ingested into the Bronze layer as timestamped JSON files, data moves to the Silver layer for cleaning and feature extraction

- Normalization: The system uses Pandas to remove textual noise, decode HTML entities, and flatten nested JSON structures from the API
- Named Entity Recognition (NER): Using the spaCy library, the pipeline identifies specific dishes and ingredients mentioned in the unstructured text.
- Sentiment Analysis: Polarity is quantified using the VADER model, which is selected for its effectiveness and low computational cost when processing short, informal culinary texts

- Trend Detection: TF-IDF-based keyword extraction is applied to detect relevant culinary recipes.

3.4 Orchestration Strategy

The entire workflow is coordinated using Apache Airflow, employing a hybrid model that combines scheduled and event-driven paradigms.

- Scheduled Ingestion (DAG 1): Operates on a weekly cadence, querying the API and RSS feed to update the Bronze layer. A weekly schedule is chosen because food trends in urban environments evolve at a rate where daily updates would introduce redundant data
- Event-Driven Processing (DAG 2): Triggered automatically by an Airflow FileSensor when new data is detected in the Bronze layer, initiating the Silver and Gold processing stages without unnecessary latency
- On-Demand Execution: The system can be triggered by the registration of a new event (e.g., a wedding), allowing the DAGs to re-execute with dynamic parameters tailored to the specific characteristics of that event

3.5 Aggregation and Visualization

In the Gold layer, the system computes summary metrics, such as top dishes by sentiment for specific event types and distributions of dietary preferences. These aggregated datasets, stored in Parquet format for performance, serve as the input for the visualization layer. The final output is an interactive dashboard built with Plotly Dash, containerized using Docker for ease of deployment, allowing catering managers to filter results by event type, trend period, or dietary needs

Chapter 4

Implementation and Results

4.1 Overview

This chapter presents the initial progress of the project, focusing exclusively on the raw data acquisition phase. At this stage, the implementation is limited to the successful consumption of the designated data sources to build a foundational Bronze layer. The results described herein reflect the collection of a sample dataset used to validate source accessibility and identify preliminary data quality challenge

4.2 Implementation Details

The current implementation consists of functional Python scripts designed to interface with the two primary data streams. For the structured source, the system utilizes the Python requests library to query the Spoonacular API, specifically targeting the /recipes/findByIngredients endpoint. For the unstructured source, a web scraping script was developed to ingest and parse the Eater NY RSS feed. All retrieved data is currently stored in its raw, untransformed state as timestamped JSON and XML files within a local directory structure. No orchestration tools or NLP libraries have been integrated into the workflow at this time

4.3 System Features and Interface

In its current state, the system lacks a graphical user interface or an interactive dashboard. The "interface" is represented by the structured output of the data collection scripts, which provide terminal-based logs of the ingestion process. The primary feature available is the ability to retrieve ingredient-specific recipe metadata and culinary article summaries from the New York City food scene

4.4 Experimental Setup

The experimental setup was conducted in a local Python environment to test the feasibility of data ingestion.

- Datasets: A controlled sample of 20 records was collected from the Spoonacular API and 20 article summaries were scraped from the Eater NY RSS feed, totaling 40 raw records

- Hardware/Software: The scripts were executed on a standard workstation using Python 3.14.2 with the requests and json libraries
- Evaluation Criteria: Success is measured by ingestion reliability, the presence of all required metadata fields (author, url, category), and the integrity of the raw JSON/XML structures.

4.5 Results

The following table summarizes the performance of the primary ingestion scripts based on the initial 40-record sample.

Table 4.1: Summary of Performance Results (Ingestion Phase)

Metric	**Proposed Method**
Spoonacular: Successful Ingestion	16
Spoonacular: Void/Unclear Structure	4
Eater NY: Successful Ingestion	20
Data Persistence Success Rate	0.90
Overall Ingestion Efficiency	80%

4.6 Validation results

Validation of the initial Bronze layer confirms that both the API and the RSS feed are viable for the project's goals. The Spoonacular API ingestion was particularly successful in retrieving multi-dimensional recipe objects that go beyond simple ingredient lists. For example, a retrieved record for "Fig and Walnut Pudding" (ID: 642780) demonstrates the system's ability to capture

- Dietary Attributes: Logical flags for vegetarian, glutenFree, and dairyFree, which are essential for fulfilling user needs regarding dietary restrictions
- Categorical Metadata: Nested arrays for dishTypes (e.g., "dessert") and diets (e.g., "paleolithic", "primal"), providing the necessary context for event-based filtering
- Nutritional Summaries: A detailed summary field that provides a textual overview of the dish's health impact

However, the inspection of these records highlights specific data quality issues that will be the primary focus of the Silver layer cleaning:

- HTML Sanitization: The summary field contains significant textual noise in the form of HTML tags (e.g., <b>, <a>) that must be stripped before Natural Language Processing can occur
- Nested Array Flattening: Key fields like diets and dishTypes are stored as nested JSON structures, necessitating normalization for efficient querying in the Gold layer

Chapter 5

Discussion

5.1 Overview

This chapter interprets the results of the initial Bronze layer implementation, which focused on the raw ingestion of culinary data from the Spoonacular API and the Eater NY RSS feed. The successful collection of a 40-record sample validates the accessibility of our heterogeneous data sources and establishes a "single version of truth" for future processing. While the system has not yet implemented advanced NLP or orchestration, the current results confirm that the necessary metadata—such as dietary attributes and editorial trends—is available to fulfill the project's long-term objectives

5.2 Interpretation of Results

The 100. The significance of the Spoonacular JSON structure (e.g., the "Fig and Walnut Pudding" example) lies in its rich metadata, which includes logical flags for vegetarian and glutenFree and detailed summary fields. Conversely, the Eater NY RSS summaries provide high-quality editorial content from the New York City food scene, but the presence of HTML entities and boilerplate noise, confirms that raw ingestion alone is insufficient for decision-making. These results emphasize the necessity of a dedicated Silver layer for data cleaning and normalization before Natural Language Processing can occur.

5.3 Addressing Project Objectives

The current implementation of the Bronze layer provides the data foundation required to meet the project's user stories:

- The ingestion process has successfully captured dish names and editorial descriptions from Eater NY . While sentiment analysis is not yet active, the availability of this textual content is the prerequisite for identifying trending and well-perceived dishes.
- The Spoonacular API response includes specific diets arrays and dietary attributes, such as dairyFree and paleolithic, which directly support the objective of providing specialized recommendations for guests with allergies or religious requirements.
- The Eater NY source provides pubDate and category fields for every article summary . This metadata will allow the system to organize insights into temporal trend windows, fulfilling the catering manager's need for weekly updates.

5.4 Limitations and Scalability

A significant limitation of the current stage is that the system relies on editorial content from Eater NY, which reflects a professional bias rather than direct, unfiltered user opinions. Additionally, the initial sample of 20 records per source is insufficient for detecting broad culinary trends in a city as large as New York. To ensure scalability, the system must transition from local execution to Apache Airflow orchestration, enabling automated weekly ingestion and the ability to process thousands of records using a Medallion Architecture .

5.5 Comparison with Related Work

The use of "event keywords" in our metadata aligns with the Context-Aware Recommender Systems (CARS) literature, which argues that traditional 2D (User-Item) matrices are insufficient for situational needs like weddings or corporate dinners. Our planned use of the VADER model for the Silver layer is supported by research identifying it as a parsimonious and effective choice for short, noisy social media and web texts. Furthermore, the project's focus on multidisciplinary drivers—combining internal sensory data (ingredients) with external social factors (trends)—mirrors established conceptual models of food choice.

5.6 Potential Improvements

Future work will focus on hardening the Bronze layer and implementing the subsequent transformation stages

Chapter 6

Conclusions

6.1 Summary

This project aimed to transform heterogeneous culinary and editorial data sources into actionable insights for professional catering managers and event planners in New York City. By bridging the gap between available recipe data and shifting public perception, the system supports a move toward data-driven menu curation.

6.2 Conclusions

Based on the results and discussion, the following conclusions are drawn:

- The feasibility of a data-driven approach to menu recommendation has been demonstrated through the successful ingestion and storage of heterogeneous data in a unified raw format.
- The combination of structured recipe information and unstructured editorial trends provides a comprehensive dual perspective (factual and perceptual) that is necessary for context-aware recommendation.
- The Medallion Architecture and proposed orchestration via Apache Airflow offer a scalable and structured framework to handle the identified technical challenges of data transformation and NLP integration.
- The system successfully captures the necessary data foundation to address the strategic and commercial needs of NYC event planners, moving menu design beyond simple intuition

6.3 Recommendations for Future Work

The following directions are recommended for future work:

- Complete the Transformation Pipeline: Implement the Silver and Gold processing layers, deploying spaCy for entity extraction and VADER for sentiment polarity scoring to convert raw records into refined insights.
- Expand Data Coverage: Incorporate direct user-generated reviews from platforms like restaurant forums or specialized review sites to mitigate the editorial bias inherent in RSS feeds.

- Advanced Modeling and Analytics: Transition from simple keyword matching to topic modeling or supervised classification to capture more complex relationships within event contexts and improve recommendation precision.
- Enhanced Dashboard Functionality: Develop advanced visualization features, such as temporal trend heatmaps and automated alerts, to support real-time, event-specific decision-making for professional users.

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